More Lock Monitoring With pg_locks

The pg_locks[17] system view provides current information about held locks. In MVCC, all transactions use either a shared or exclusive lock of database resources like a table or rows of data.

You observed lock activity focused on table level locks and concurrent modifications to table structures. Now you'll focus on errors with concurrency from DML operations.

Besides DDL changes like ALTER TABLE that use an exclusive lock, Autovacuum uses a SHARE UPDATE EXCLUSIVE lock mode. This lock type means that it is both an exclusive lock and a shared lock. Access for reads is Shared, but is locked for updates.

When an UPDATE occurs to a row, the row is locked exclusively during the update. Concurrent reads of the row are possible.

SELECT ... FOR UPDATE adds additional protection when selecting rows being updated (See: *Selecting for Share and Update in PostgreSQL*[18]).

ROW SHARE LOCK locks are taken on those rows. Although rows could be selected and updated within a transaction, a concurrent transaction could modify the same rows being selected.

The solution is to open a transaction and select the rows, but add the FOR UPDATE keyword to indicate that rows will be exclusively locked for the duration of the SELECT statement.

Taking an exclusive lock on rows will block other row SELECT operations and modifications. Any concurrent readers or modifications will be in a waiting state for the SELECT ... FOR UPDATE transaction block to COMMIT or ROLLBACK.

The NOWAIT keyword can be added to the SELECT ... FOR UPDATE to prevent excessive waiting to acquire the lock on the rows to update.

FOR UPDATE is considered to create an explicit lock, similar to how the LOCK command may be used to explicitly lock a table.

Monitoring Row Locks

Set the log_min_error_statement[19] to log queries to postgresql.log that may time out due to being blocked on lock acquisition.

17. https://www.postgresql.org/docs/current/view-pg-locks.html
18. https://shiroyasha.io/selecting-for-share-and-update-in-postgresql.html
19. https://www.postgresql.org/docs/current/runtime-config-logging.html